Fold-dependent slowing in mechanistic CSD

This supplement collects manuscript-adjacent details that are useful for review but too long for the main text: the full paired geometry sweep behind the mechanistic slowdown statistics, the exact representative-run outputs underlying manuscript Table 1, the atlas-patch sanity-check figure, and a narrow physiology-anchor check against published CSD speed and delay bands. The physiology-anchor items come from the reduced biophysical validation workflow and should be read as contextual support rather than as a direct fit of the multi-ion mechanistic surface model.

Supplementary Tables

The speed bands summarized in Table S3 and Fig. S1 are the same published experimental and clinical ranges discussed in the main manuscript. In the present supplement they are included only as a narrow contextual check that accepted scalar and tensor validation cases remain within published speed regimes while preserving the tensor-faster-than-scalar ordering.

Table S1: Full paired geometry sweep for the mechanistic surface electrodiffusion model. Each row compares the same geometry with dipole alignment disabled or enabled. The flat control is included as an internal specificity check and shows zero slowing.

Depth (mm)	σ (mm)	Severity	Baseline speed (mm/min)	Dipole speed (mm/min)	Slowdown (mm/min)	Baseline delay (s)	Dipole delay (s)	Delay increase (s)	Baseline $\lvert V_e \rvert$ (mV)	Dipole $\lvert V_e \rvert$ (mV)
0.0	1.5	0.000	3.623	3.623	0.000	64.3	64.3	0.0	19.43	19.43
1.2	1.1	1.091	2.355	2.259	0.096	135.1	141.5	6.4	21.88	21.81
1.2	1.5	0.800	2.368	2.239	0.128	130.3	139.9	9.6	21.54	21.68
1.2	1.9	0.632	2.415	2.337	0.078	124.8	128.9	4.1	21.09	21.24
1.8	1.1	1.636	2.603	2.485	0.118	135.2	141.8	6.6	22.11	22.01
1.8	1.5	1.200	2.554	2.415	0.140	130.0	139.4	9.3	21.65	21.80
1.8	1.9	0.947	2.556	2.439	0.117	124.5	130.4	5.9	21.14	21.03
2.4	1.1	2.182	2.923	2.783	0.140	135.1	142.2	7.0	22.29	22.14
2.4	1.5	1.600	2.802	2.646	0.156	129.8	138.6	8.8	21.75	21.90
2.4	1.9	1.263	2.736	2.661	0.075	124.1	127.5	3.4	21.23	21.37
3.0	1.1	2.727	3.287	3.129	0.158	135.1	142.2	7.0	22.44	22.18
3.0	1.5	2.000	3.071	2.833	0.238	129.5	141.8	12.3	21.85	21.66
3.0	1.9	1.579	2.898	2.816	0.082	123.6	126.8	3.2	21.32	21.45

Table S2: Exact representative-run outputs underlying manuscript Table 1 for the folded surface used in the main manuscript (fold depth 2.4 mm, $\sigma = 1.5$ mm). Values are copied directly from the saved final representative run, not from a rerun. The full coupled run is retained only as an exploratory sensitivity analysis because vascular feedback changes the excitability regime.

Case	dt (s)	Speed (mm/min)	E1 arrival (s)	E2 arrival (s)	Delay (s)	Max $\lvert V_e \rvert$ (mV)	Max K_e (mM)	Max swelling	Min oxygen	Min perfusion
Baseline mechanistic	0.04996995735874618	2.5582349901822656	30.681553818270157	161.20308243931518	130.52152862104504	18.987306193669685	35.6329071567077	1.4998735073093261	1.0	0.6
Dipole-aligned mechanistic	0.04996995735874618	2.475787705816325	30.681553818270157	165.3505889000911	134.66903508182097	19.50089527429308	35.61903013535566	1.4998731422865321	1.0	0.6
Full coupled sensitivity	0.04996995735874618	3.546505127010437	30.431704031476425	124.52513373799549	94.09342970651906	12.95984978092811	32.865620644529656	1.4998601120671369	0.2	0.25

Table S3: Accepted-candidate physiology anchor from the reduced biophysical validation sweep. This table is included as a contextual range check against published CSD speed and delay bands, not as a direct validation of the multi-ion mechanistic surface model.

Metric	Range (mm/min)	Mean	Within 2.0–5.0	Within 1.7–9.2	Within 2.5–4.5
Control	3.000–3.000	3.000	63/63	63/63	63/63
Scalar	2.606–2.750	2.635	63/63	63/63	63/63
Tensor	2.683–2.792	2.706	63/63	63/63	63/63
Tensor - scalar	0.042–0.077	0.071	–	–	–
Scalar downstream delay (s)	0.0–16.2 s	12.8 s	Within 10–30 s band: 54/63		

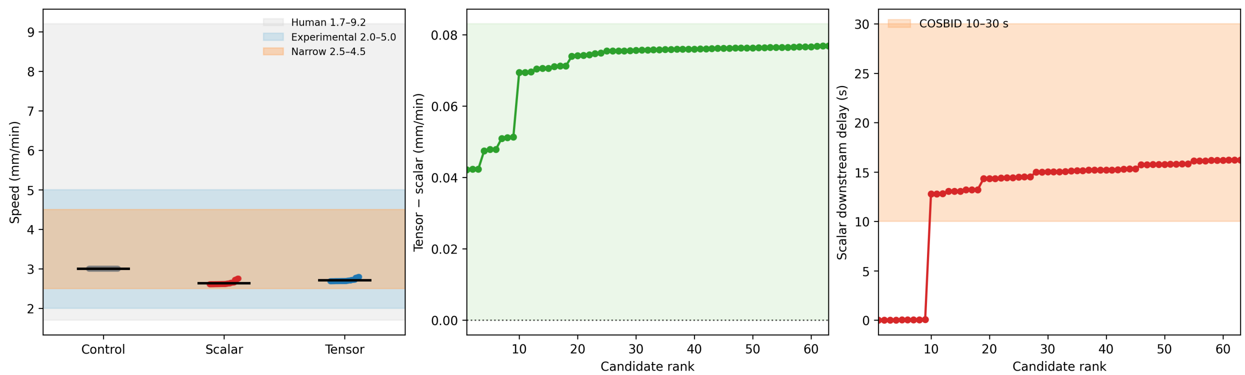

Figure S1: Physiology-anchor summary for the accepted reduced-biophysical validation cases. Shaded or annotated ranges denote the same broad published CSD speed and delay bands discussed in the main manuscript. The figure is included as a contextual supplement and should not be interpreted as a direct waveform-level validation of the mechanistic surface electrodiffusion model.

Supplementary Figures

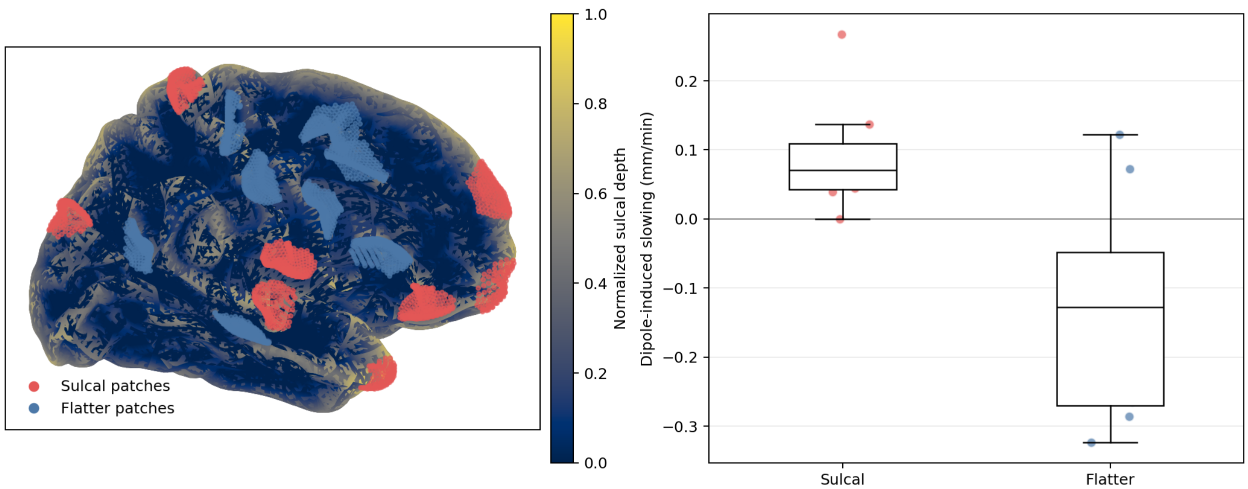

Figure S2: Atlas-based multi-patch sanity check used as a geometry-specific supplement to the main mechanistic results. Left, the fsaverage6 left-hemisphere midthickness surface with the $N = 8$ selected sulcal and $N = 8$ flatter matched patches overlaid on the normalized sulcal-depth map. Right, the distributions of dipole-induced propagation slowing for each patch category. The consistent positive slowdown on sulcal patches supports the main mechanistic results, whereas flatter patches (which still possess local structural curvature) yield inverted effects. This figure remains supplementary because it functions as an atlas-based sign-check rather than a full subject-specific human validation study.